

# A Synthetic Benchmark of Strain-Hardening-Inspired Features for Discriminating Sheet Positions (Edge/Middle/Center)

## Abstract

Position-dependent mechanical variability (e.g., Edge vs Center) is frequently observed in sheet materials due to heterogeneous processing histories and microstructural differences. For practical diagnostics and quality control, compact and interpretable descriptors are often preferable to opaque curve-level models. This study presents a reproducible synthetic benchmark to examine whether strain-hardening-inspired descriptors can discriminate sheet positions (Edge, Middle, Center). Instead of operating on raw stress-strain curves, we generate a tabular feature dataset that emulates plausible trends in true stress and strain hardening rate (SHR) at selected strain levels, along with summary descriptors including maximum SHR, early-regime mean SHR, proxy uniform true strain, and the Hollomon exponent. Class-dependent offsets and stochastic noise are applied to represent systematic position signatures while maintaining realistic variability. Using the resulting dataset (18 groups  $\times$  3 positions;  $N = 54$ ), we compare feature distributions via box-and-strip plots and summarize class-wise mean profiles using both raw-scale and z-score normalized bar plots. The results show consistent and interpretable shifts among positions across several SHR-related descriptors, indicating that a compact set of physically meaningful features can encode position-dependent mechanical signatures within the designed benchmark. This work provides a transparent baseline protocol for feature engineering and visualization prior to applying the same pipeline to experimental tensile datasets.

## 1. Introduction

Mechanical properties of sheet materials can vary across positions such as Edge, Middle, and Center. These differences may originate from spatially varying strain histories, residual stresses, or microstructural heterogeneity introduced during manufacturing. While curve-level machine learning can be effective, engineers often prefer compact and interpretable descriptors for rapid screening, visualization, and physically grounded interpretation.

Strain-hardening behavior is especially informative because it reflects deformation mechanisms such as dislocation storage and dynamic recovery. Common strain-hardening-related descriptors include the strain hardening rate

$\theta = d\sigma/d\varepsilon$  and the Hollomon exponent  $n$  from  $\sigma = K\varepsilon^n$

Motivated by these considerations, this study proposes a reproducible synthetic benchmark to evaluate whether a small set of strain-hardening-inspired features can discriminate sheet positions in an interpretable way.

Scope and contribution. The goal is to provide a transparent baseline protocol for feature engineering and visualization. The dataset used here is synthetic by construction; therefore, conclusions are restricted to behavior within the benchmark, and real-data generalization is treated as future work.

## 2. Methods

### 2.1 Dataset and outputs

A synthetic feature dataset was generated and saved as `features_all.csv` with  $N = 54$  samples ( $18 \text{ groups} \times 3 \text{ positions: Edge/Middle/Center}$ ). Each row represents one sample. The columns include numerical descriptors (“features”), and categorical identifiers:

label: class label (Edge/Middle/Center)

group: paired group identifier (e.g., `synth_001`)

file, sheet: tracking fields (position metadata)

Three figures were produced:

Figure 1: `stylish_boxplots.png` (distribution comparison by class)

Figure 2a: `stylish_mean_barplot.png` (class-wise means, raw scale)

Figure 2b: `stylish_mean_barplot_zscore.png` (class-wise means, z-score normalized)

### 2.2 Feature definitions

The dataset contains strain-hardening-inspired descriptors grouped as follows:

(i) True stress at fixed true strain

`ts@0.02`, `ts@0.05`, `ts@0.10`, `ts@0.15`, `ts@0.20`

→ true stress evaluated at selected true strain points.

(ii) Strain hardening rate (SHR) at fixed true strain

`shr@0.02`, `shr@0.05`, `shr@0.10`, `shr@0.15`, `shr@0.20`

→  $d\sigma/d\varepsilon$  evaluated at the same strain points.

(iii) Summary hardening metrics

`shr_max`, `shr_max_strain`, `shr_min`, `shr_min_strain`, `shr_mean_0_0.05`

→ peak/minimum hardening rate and their strain locations; early-regime average hardening.

(iv) Strength and integral summaries

`ts_max`, `auc_ts`, `auc_shr`

→ maximum true stress; area-under-curve summaries for stress and SHR over the defined strain range.

(v) Proxy ductility and Hollomon exponent

`uniform_true_strain_est`, `hollomon_n_0.02_0.15`

→ a proxy for uniform true strain and a proxy estimate of  $n$  over 0.02-0.15.

### 2.3 Synthetic generation logic and reproducibility

Features were generated with class-dependent offsets such that Edge, Middle, and Center exhibit systematic differences in strength/hardening/ductility proxies, with added random noise to emulate variability. A fixed seed (seed = 42) was used to ensure reproducibility. Because the dataset is synthetic and separation is partly embedded by design, the benchmark is intended for feature engineering evaluation and visualization protocol testing rather than final validation on experimental data.

### 2.4 Visualization

Distribution comparison (Fig. 1): Box-and-strip plots for selected key features.

Mean profile comparison (Fig. 2a): Class-wise mean bar plot on raw feature scale.

Scale-independent comparison (Fig. 2b): Z-score normalization applied per feature across all samples, followed by class-wise mean bar plot. This resolves visualization bias caused by differing feature magnitudes.

| Category         | Item          | Setting / Value                      | Notes                         |
|------------------|---------------|--------------------------------------|-------------------------------|
| Data type        | Generation    | Synthetic tabular feature generation | Not experimental tensile data |
| Reproducibility  | Random seed   | seed = 42                            | Deterministic regeneration    |
| Sample structure | No. of groups | n_synthetic_files = 18               | synth_001to synth_018         |
| Sample structure | Classes       | Edge / Middle / Center               | labelcolumn                   |
| Sample count     | Total samples | 54 rows (= 18 × 3)                   | One row per sample            |
| Feature (stress) | Strain points | 0.02, 0.05, 0.10, 0.15, 0.20         | ts@x.xx                       |

|                    |                      |                                               |                             |
|--------------------|----------------------|-----------------------------------------------|-----------------------------|
|                    |                      |                                               |                             |
| Feature (SHR)      | Strain points        | 0.02, 0.05, 0.10, 0.15, 0.20                  | shr@x.xx                    |
| Feature (summary)  | Peak/min SHR         | shr_max, shr_min+ locations                   | *_strain                    |
| Feature (summary)  | Early mean SHR       | shr_mean_0_0.05                               | Early hardening regime      |
| Feature (integral) | AUC summaries        | auc_ts, auc_shr                               | Over defined strain range   |
| Feature (proxy)    | Ductility / exponent | uniform_true_strain_est, hollomon_n_0.02_0.15 | Proxy estimates             |
| Class design       | Offsets              | Class-dependent feature offsets               | Emulate position signatures |
| Variability        | Noise                | Random noise added per feature                | Emulate dispersion          |
| Outputs            | Dataset              | features_all.csv                              | In ML_results_stylish       |
| Outputs            | Figures              | Boxplot + mean plots                          | Fig.1, Fig.2a, Fig.2b       |

Table 1. Summary of Experimental/Data (Synthetic) Conditions

### 3. Results

#### 3.1 Distributional separation across positions (Figure 1)

Figure 1 shows systematic shifts among Edge/Middle/Center for multiple descriptors. Several strain-hardening-related features display the clearest class separation with relatively limited overlap, including:

shr\_max and shr\_mean\_0\_0.05 (peak/early hardening behavior),

shr@0.15 (hardening behavior at moderate strain),

hollomon\_n\_0.02\_0.15 and uniform\_true\_strain\_est (hardening-law/ductility proxies),

auc\_shr (integral hardening summary).

In comparison, a single stress-point descriptor such as ts@0.15 shows weaker separation and more overlap. This indicates that, within the benchmark, hardening-related descriptors encode position signatures more distinctly than a single strength snapshot.

위치별 주요 재료 특성 분포 비교

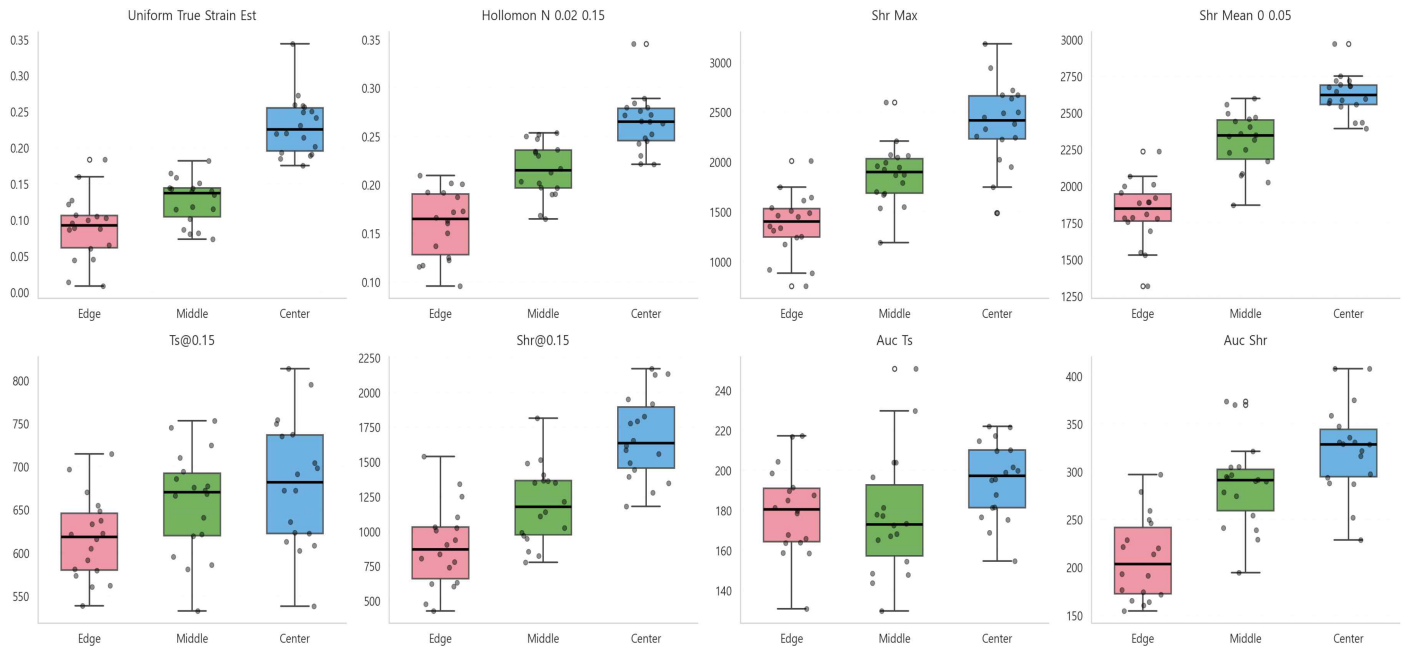


Figure 1. Distributional separation across positions

#### 3.2 Mean profile comparison (Figures 2a and 2b)

Figure 2a confirms the position-dependent ordering for large-magnitude descriptors (particularly SHR-related metrics), where class-wise mean differences are visually apparent. However, because features exist on very different numerical scales, smaller-magnitude descriptors (e.g., uniform\_true\_strain\_est, hollomon\_n\_0.02\_0.15) appear compressed on the raw-scale plot.

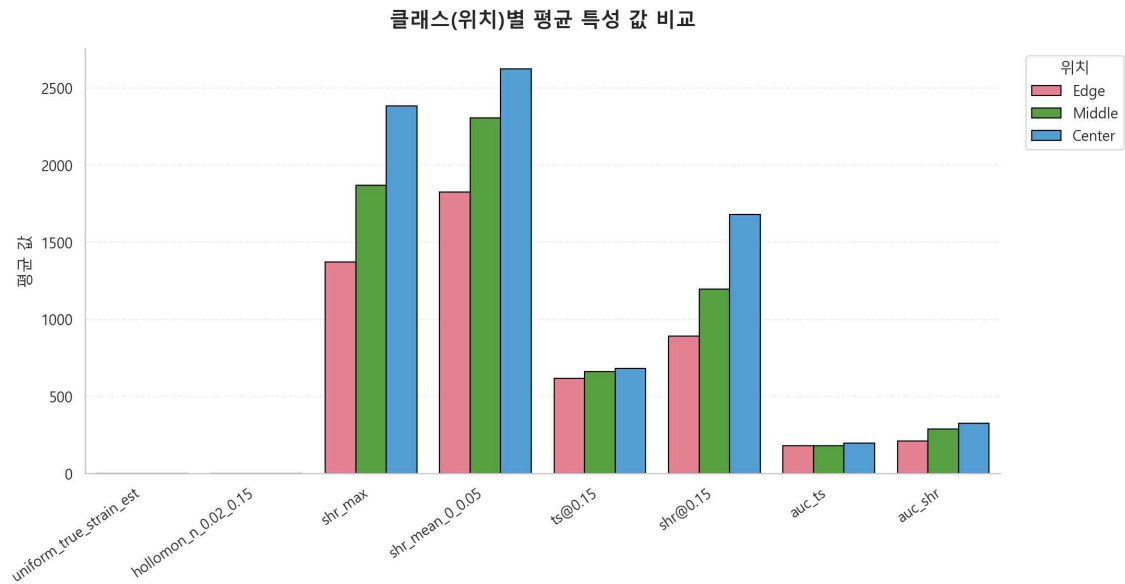


Figure 2a. Class-wise mean feature values on the original scale.

Figure 2b addresses this by using z-score normalization per feature before computing class-wise means. The normalized plot enables fair cross-feature comparison and confirms that even smaller-scale descriptors follow consistent position-dependent trends.

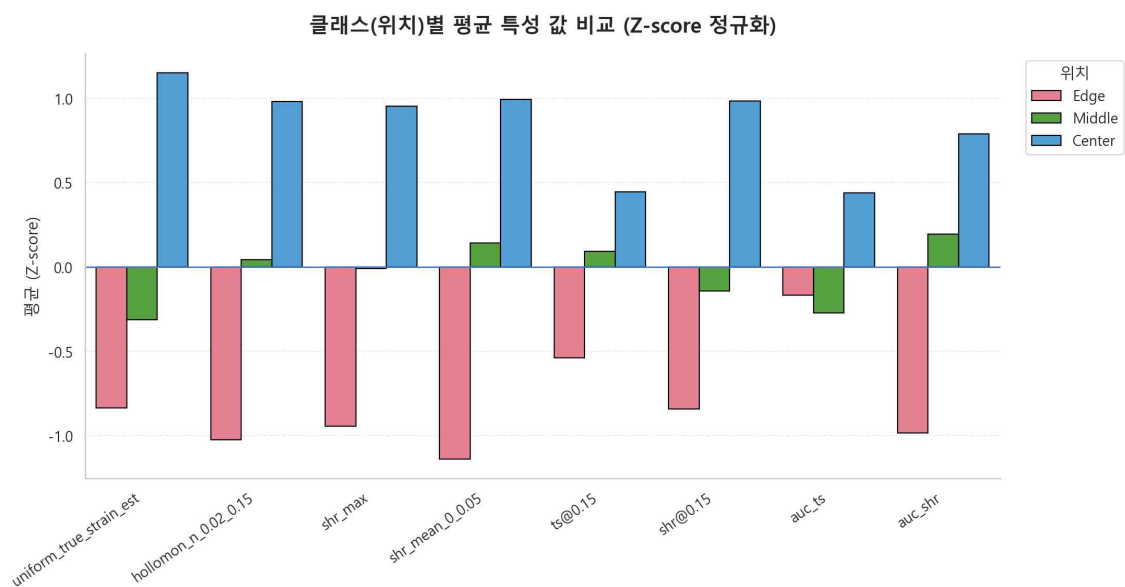


Figure 2b. Class-wise mean feature values after per-feature z-score normalization for scale-independent comparison.

## 4. Discussion

This benchmark demonstrates that strain-hardening-inspired descriptors can encode position-dependent signatures in an interpretable manner. Compared with single-point stress measures, hardening-related descriptors (peak hardening, early-regime hardening, moderate-strain hardening, and hardening-law proxies) provide clearer separation among Edge/Middle/Center within the designed benchmark. This supports the practical idea that deformation-mechanism-informed descriptors may be more diagnostic than strength alone.

Limitations. The dataset is synthetically generated with class-dependent offsets; therefore, observed separability reflects behavior within the benchmark and does not guarantee generalization to experimental tensile data. Additionally, the current study focuses on visualization and protocol development rather than downstream model training and performance metrics.

Future work. The next step is to apply the same feature definitions to experimentally measured stress-strain curves, quantify sensitivity to smoothing/derivative estimation, and validate robustness across processing conditions (e.g., heat-treatment time/temperature or different sheet batches).

## 5. Conclusion

We proposed a reproducible synthetic benchmark using strain-hardening-inspired features to discriminate sheet positions. The results show consistent and interpretable shifts among Edge/Middle/Center across multiple hardening-related descriptors. The released dataset and visualization protocol provide a transparent baseline prior to extension to experimental tensile datasets.

### Figure Captions

Figure 1. Distribution comparison of key strain-hardening-inspired descriptors across positions (Edge/Middle/Center) using box-and-strip plots.

Figure 2a. Class-wise mean comparison on raw feature scale.

Figure 2b. Class-wise mean comparison after per-feature z-score normalization to enable scale-independent comparison.